PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Instruments measuring the quality of life among people living with type-2 diabetes mellitus in India: A systematic review protocol
AUTHORS	Demla, Chetna; Thomas, Anns; Jose, Jomol; Joshy, Angela; Hrishikesh, M; Rajendran, Ambigai; Parsekar, Shradha

VERSION 1 – REVIEW

REVIEWER	Jill Carlton University of Sheffield United Kingdom I am the lead author of a submitted manuscript which examines instruments that that been used to measure the impact of hypoglycaemia upon quality of life, using COSMIN methodology. The review was undertaken as part of the Hypoglycaemia REdefining SOLutions for better liVEs (Hypo-RESOLVE) project, an international collaboration of clinicians, scientists, industry partners and people with diabetes [19. de Galan BE, McCrimmon RJ, Ibberson M, Heller SR, Choudhary P, Pouwer F, Speight J, Carlton J, Pieber TR, Rosilio M, Tack CJ, Müllenborn M for the Hypo-RESOLVE Consortium. Reducing the burden of hypoglycaemia in people with diabetes through increased understanding: design of the Hypoglycaemia REdefining SOLutions for better liVEs (Hypo-RESOLVE) project. Diabetic Med 2020; 37(6): 1066-1073.].
REVIEW RETURNED	27-Aug-2020

GENERAL COMMENTS	Thank you for the opportunity to comment upon the protocol. Please find a couple of comments for consideration. 1. Whilst I do appreciate that there is a difference between PROSPERO registration and a systematic review protocol, I wonder whether there is a need for the protocol to be published given that the information provided within the protocol is no different to that already shown on PROSPERO? For example, the information on participants and outcome are the same, as is data extraction, risk of bias assessment, and description of analysis and synthesis. What does the protocol add that the PROSPERO documentation lacks? 2. The search will identify many instruments that have been used to measure the impact of diabetes upon QoL. You may wish to consider the addition of some further criteria to negate the large numbers of PROMs that you will identify. For example, how are you defining quality of life? What if a PROM has been used to measure one construct of quality of life (i.e. depression)? Will that be included within the review? Are you considering all measures of

	quality of life (including generic measures such as EQ-5D or SF-36)? Or do you plan only to include diabetes-specific measures? 3. Whilst patient and public involvement has not been undertaken for designing or writing the protocol, are there plans for patient and public involvement within the review itself? If not, why not?

REVIEWER	S BELLARY Aston University Birmingham, UK
REVIEW RETURNED	26-Nov-2020

GENERAL COMMENTS	The manuscript is a description of protocol for undertaking a systematic review of QOL life instruments in Type 2 diabetes patients of Indian origin. Given the rising number of people with type 2 diabetes in the Indian sub-continent, the proposed systematic review would be of relevance in identifying key instruments to measure the QOL appropriate for the population. While the proposed methodology is broadly consistent with other systematic reviews, I have a few comments to make which I have listed below.Introduction Line 67 : Type 2 diabetes is not just Insulin resistance but a combination of beta cell dysfunction and insulin resistance Line 70: Check continuity of the sentence Line 74: Rather say perceived, please say it is a worldwide epidemic Line 78 : Nevertheless 'India Today 2019'. Not clear what this means Line 85/86 : 'especially those who live in developing countries…' . Please note that QOL is an issue in all countries not just developing countries . Line 97: I suppose you mean the 'negative' impact of diabetes on QOL Line 110: 'Several studies have been conducted ……' Here I see no references provided even though the sentence mentions several studies. Also, there is no description of the nature of these studies, the populations studied and the instruments of interest. Clearly, the main purpose of the review is to answer some of these but even so, it would be useful to provide a context here to make the justification for undertaking the systematic review. Line 114/115: This sentence is a bit vague . While referring to specific factors, it would be good to give a few examples of them. General comment: I would have loved to see a more convincing argument about the need for this review. Some detail about the instruments studied and why and how are these different in the Indian context would have been good. Methods The methods described here are satisfactory and address the key elements of the review.

	The one thing that is clearly missing is the detail of any statistical analysis and meta-analysis they propose to undertake. Other Although this paper is mainly to publish the protocol, it ends rather abruptly. It would be good to provide the reader with an overview of how this methodology differs (or is similar to)from other similar systematic reviews. Similarly, some discussion about the HRQOL in diabetes and impact of other conditions on HRQOL in the same population would be good. These are only a few suggestions but the authors can expand on these themes.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1

1. Whilst I do appreciate that there is a difference between PROSPERO registration and a systematic review protocol, I wonder whether there is a need for the protocol to be published given that the information provided within the protocol is no different to that already shown on PROSPERO? For example, the information on participants and outcome are the same, as is data extraction, risk of bias assessment, and description of analysis and synthesis. What does the protocol add that the PROSPERO documentation lacks?

Reply: Thank you for raising this concern. PROSPERO is a registration platform, it merely indicates that a requirement of registration has been fulfilled. It is neither peer-reviewed nor a publication. It provides a minimum set of statements required to register and is not an exhaustive record of the background, need of the study, and methodology. We had used tables to strengthen the details in manuscript, which were not in PROSPERO. We agree that some of the detail in PROSPERO and manuscript submitted to the journal were same, however, in the current form of manuscript, we have provided more details, to demonstrate that the publication is a value addition, not a replacement. We would want the rigorous peer-review of our protocol so as to have a citable validation to the researchers.

2. The search will identify many instruments that have been used to measure the impact of diabetes upon QoL. You may wish to consider the addition of some further criteria to negate the large numbers of PROMs that you will identify. For example, how are you defining quality of life? What if a PROM has been used to measure one construct of quality of life (i.e. depression)? Will that be included within the review? Are you considering all measures of quality of life (including generic measures such as EQ-5D or SF-36)? Or do you plan only to include diabetes-specific measures?

Reply: We do not impose any restriction on the definition of QoL. Studies measuring specific construct of QoL if has been measured are eligible to be included. We are including any types of QoL, irrespective whether it is generic or diabetes specific, provided it should have been conducted among resident Indian with diabetes. We agree with the reviewer that we will end-up including a large no. of PROM therefore have refined the inclusion/exclusion criteria. Kindly refer to inclusion/exclusion criteria on pages 7-8 of manuscript and furthermore screening protocol in table 2.

3. Whilst patient and public involvement has not been undertaken for designing or writing the protocol, are there plans for patient and public involvement within the review itself? If not, why not?

Reply: We had not initially planned to involve patients but considering the importance of stakeholders in research we will explore the possibility of including the stakeholders at the final stage of review. We have reported how we will do this in main manuscript. Kindly refer to page 18 of manuscript.

Reviewer 2

The manuscript is a description of protocol for undertaking a systematic review of QOL life instruments in Type 2 diabetes patients of Indian origin. Given the rising number of people with type 2 diabetes in the Indian sub-continent, the proposed systematic review would be of relevance in

identifying key instruments to measure the QOL appropriate for the population. While the proposed methodology is broadly consistent with other systematic reviews, I have a few comments to make which I have listed below.

Introduction

1. Line 67: Type 2 diabetes is not just Insulin resistance but a combination of beta cell dysfunction and insulin resistance

Reply: We have edited the statement as suggested. The statements reads, '"Type-2 diabetes mellitus" …… occurs when body is unable to make required amount of insulin or doesn't use insulin appropriately. It is caused due to beta cells dysfunction and insulin resistance.' Please refer page 4, first paragraph of Introduction.

2. Line 70: Check continuity of the sentence.

Reply: We have edited the statement, 'As a result of ignorance of these initial symptoms/signs, diabetes may be diagnosed many years after onset…….' Page 4, first paragraph.

3. Line 74: Rather say perceived, please say it is a worldwide epidemic

Reply: We have removed the word 'perceived' from the statement. Refer line 77. We have also removed the same word from abstract.

4. Line 78: Nevertheless 'India Today 2019'. Not clear what this means

Reply: We have edited the statement, 'Nevertheless, India already has the maximum estimated number of people diagnosed with diabetes in the world'. Line 80-81.

5. Line 85/86: 'especially those who live in developing countries…'. Please note that QOL is an issue in all countries not just developing countries.

Reply: We agree with the comments and therefore, have edited the statement. Refer line 89.

6. Line 97: I suppose you mean the 'negative' impact of diabetes on QOL

Reply: Thanks for pointing this. Yes it is negative impact. We have added the word as suggested. Line 101.

7. Line 110: 'Several studies have been conducted ……' Here I see no references provided even though the sentence mentions several studies. Also, there is no description of the nature of these studies, the populations studied and the instruments of interest. Clearly, the main purpose of the review is to answer some of these but even so, it would be useful to provide a context here to make the justification for undertaking the systematic review.

Reply: Thank you for pointing this. We, now, have cited some of the studies conducted in India and have provided important findings and other details of these studies. Lines 115-122.

8. Line 114/115: This sentence is a bit vague. While referring to specific factors, it would be good to give a few examples of them.

Reply: We, now, have reported some of the factors that we think are important considering the Indian context that may influence quality of life. Refer lines 123-138

Methods

9. The methods described here are satisfactory and address the key elements of the review. The one thing that is clearly missing is the detail of any statistical analysis and meta-analysis they propose to undertake.

Reply: Our intension is to carry out the narrative synthesis and meta-analysis for the type of review that we want to undertake is not appropriate. We have provided the justification for not performing statistical pooling. Refer lines 240-245.

Other

10. Although this paper is mainly to publish the protocol, it ends rather abruptly. It would be good to provide the reader with an overview of how this methodology differs (or is similar to) from other similar systematic reviews. Similarly, some discussion about the HRQOL in diabetes and impact of other conditions on HRQOL in the same population would be good. These are only a few suggestions but the authors can expand on these themes.

Reply: We agree with the comment raised, we followed the journal guidelines that is why the abrupt end. However, the detailed description of the significance of this research can be found on pages 6-7. The main highlights can also be found in article summary, page 3.

VERSION 2 – REVIEW

REVIEWER	Carlton, Jill University of Sheffield, School of Health and Related Research
REVIEW RETURNED	06-Jan-2021

GENERAL COMMENTS	Thank you for the opportunity to review the revised manuscript. I have made some comments on the document attached. I am not sure if I am misinterpreting your description of how you propose to apply the COSMIN methods to your review, or whether you are aware of how to apply the COSMIN methods correctly. At present the COSMIN manuals are not referred to within the manuscript, and these do fully detail how to undertake the review process. The areas of concern relate to the searches themselves (no mention of the PROMs filter developed by Terwee et al, 2009), and little detail on the rating of the psychometric properties. This may of course be difficult to incorporate due to word limits, however you can refer to the manuals and inform the reader that you will be adhering to the steps outlined in those documents (two manuals, one specifically for content validity, and one for SR of PROMs). You may wish to refer to (and read) other reviews that have been published using the COSMIN approach. 1. Chiarotto A, Ostelo RW, Boers M, Terwee CB. A systematic review highlights the need to investigate the content validity of patient-reported outcome measures for physical functioning in patients with low back pain. J Clin Epidem 2018; 95: 73-93. 2. Craxford S, Deacon C, Myint Y, Ollivere B. Assessing outcome measures used after rib fracture: A COSMIN systematic review. Injury 2019; 50(11): 1816-1825. 3. Rezai M, Kolne K, Bui S, Lindsay S. Measures of Workplace Inclusion: A Systematic Review Using the COSMIN Methodology. J Occup Rehabil 2020; 30: 420-454. https://doi.org/10.1007/s10926-020-09872-4 4. Ortega-Avila AB, Cervera-Garvi P, Ramos-Petersen L, Chicharro-Luna E, Gijon-Nogueron G. Patient-reported outcome measures for patients with diabetes mellitus associated with foot and ankle pathologies: A systematic review. J Clin Med 2019;8(2):146. 5. Powell PA, Carlton J, Buckley Woods, H, Mazzone P. Measuring quality of life in Duchenne muscular dystrophy: a systematic review of the content and structural validity of commonly used instruments. Health Qual Life Outcomes 2020;18:263.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 1
Thank you for the opportunity to review the revised manuscript. I have made some comments on the document attached.
Reply: Thank you for the constructive feedback. We are grateful for suggesting the edits.

I am not sure if I am misinterpreting your description of how you propose to apply the COSMIN methods to your review, or whether you are aware of how to apply the COSMIN methods correctly. At present the COSMIN manuals are not referred to within the manuscript, and these do fully detail how to undertake the review process. The areas of concern relate to the searches themselves (no mention of the PROMs filter developed by Terwee et al, 2009), and little detail on the rating of the psychometric properties. This may of course be difficult to incorporate due to word limits, however you can refer to the manuals and inform the reader that you will be adhering to the steps outlined in those documents (two manuals, one specifically for content validity, and one for SR of PROMs).
Reply: Thank you for the suggestions. Yes, we agree with the referee, we had faintly reported COSMIN study without providing detailed description. However, in the current protocol we have incorporated the suggestions and have provided detailed description, wherever applicable.
Reply: Thank you for the constructive feedback. We have appropriately reported and cited the COSMIN methodology in the manuscript.

You may wish to refer to (and read) other reviews that have been published using the COSMIN approach.
1. Chiarotto A, Ostelo RW, Boers M, Terwee CB. A systematic review highlights the need to investigate the content validity of patient-reported outcome measures for physical functioning in patients with low back pain. J Clin Epidem 2018; 95: 73-93.
2. Craxford S, Deacon C, Myint Y, Ollivere B. Assessing outcome measures used after rib fracture: A COSMIN systematic review. Injury 2019; 50(11): 1816-1825.
3. Rezai M, Kolne K, Bui S, Lindsay S. Measures of Workplace Inclusion: A Systematic Review Using the COSMIN Methodology. J Occup Rehabil 2020; 30: 420-454. https://doi.org/10.1007/s10926-020-09872-4
4. Ortega-Avila AB, Cervera-Garvi P, Ramos-Petersen L, Chicharro-Luna E, Gijon-Nogueron G. Patient-reported outcome measures for patients with diabetes mellitus associated with foot and ankle pathologies: A systematic review. J Clin Med 2019;8(2):146.
5. Powell PA, Carlton J, Buckley Woods, H, Mazzone P. Measuring quality of life in Duchenne muscular dystrophy: a systematic review of the content and structural validity of commonly used instruments. Health Qual Life Outcomes 2020;18:263.
Reply: Thank you so much for sharing these published reviews, it was really helpful to understand the methodology better.

Specific comments
1. (line 26, Comment JC1) Identify may be better
Reply: The word “Identify” is added to the sentence as mentioned in the comment. Please refer line no. 25.
2. (line 27, Comment JC2) You may wish to consider explaining what is meant by dimensions, not all readers will be familiar with this term. I am not sure what you are referring to, do you mean what concepts/themes of QoL the instruments measure?
Reply: As per the suggestion, the word “Dimensions” has been replaced by “Domains”. Quality of life is a multi-dimensional construct and thus may include domains such as psychological or physical components. Please refer line no. 26 in abstract and line no. 128, 156.
3. (line 29, Comment JC3) I would add “using COSMIN methodology” to the end of this sentence. COSMIN methodology is regarded as being robust and evidenced based, by adding this detail it informs the reader that you will be applying recognized approach(es).
Reply: The statement has been rephrased according to the suggestion. Please refer line nos. 28, 29.
4. (line 32, Comment JC4) Could rephrase? Predetermined inclusion/exclusion criteria will be applied to the searches, which include…..”
Reply: Sentence has been rephrased according to the suggestion. Please refer line 33.
5. (line 36, Comment JC5) The COSMIN checklist is different to what you are doing in the review though. You are using COSMIN methodology to assess the psychometric properties of identified

instruments, and part of the COSMIN approach incorporates assessment of the methodological quality of the included studies. You can (should you wish) also choose to apply the COSMIN checklist to your own review.
Reply: Thank you for raising this concern. The methodological quality of included studies will be assessed using the COSMIN Risk of Bias checklist. Yes, the quality of our systematic review itself will be evaluated using COSMIN checklist. We have made the necessary changes in the manuscript file. Please refer line nos. 37, 38 in abstract and 203-26.
6. (line 50, Comment JC6) You are not using COSMIN methods to define the psychometric properties. You are using COSMIN methods to assess the psychometric properties of the instruments you identify. The sentence would benefit from rewording.
Reply: Rewording has been made as suggested. Refer lines 49, 50.
7. (line 51, Comment JC7) Is this relevant to include in this section?
Reply: We have replaced the third point of article summary. Please refer lines 51-52.
8. (line 75, Comment JC8) I am not sure ignorance is the right word here – individuals may be aware of the initial signs and symptoms, but they can be linked to many health conditions.
Reply: The word “lack of awareness” has been used in place of ignorance. Yes, we agree, there are many demand and supply-side factors that may be responsible for the delay in diagnosis of diabetes. We now have added this information. Please refer lines 74-76.
9. (line 114, Comment JC9) I am not sure what this adds – it could be summarized to cover a couple of key points.
1.Studies have taken place in India where the QoL of people with T2DM has been investigated. It is difficult to summarise the results of those studies as they used different QoL instruments.
2.The instruments used in the studies may not best capture QoL for individuals either living in India and/or living in India with diabetes (due to cultural differences).
3.Your review aims to
a.Identify what instruments have been used in studies of people living in India with T2DM
b. Critique the psychometric properties of the instruments identified in a) using established methodology (i.e. COSMIN).
Reply: Thank you so much for summarising the entire paragraph. We have accepted the suggestions. Please refer lines 111-120.
10. (line 145, Comment JC10) I am not sure what you mean by this? Are you looking at high level domains, such as psychological functioning or physical functioning? Or subdomains, such as fear or ability to self-care?
Reply: The domains in this context refers to psychological or physical components of the tool. We have now explicitly stated this at appropriate places. Please refer line 128.
11. (line 165, Comment JC11) This would benefit from rewording. The primary outcome is to identify all self-report instruments that have been used to assess quality of life in studies of individuals from India with T2DM.
Reply: The statement has been rephrased. Please refer lines 151-152.
12. (line 169, Comment JC12) I am not sure I fully understand you here. Are you saying that you will include all instruments even if they have not been validated for use in India? I.e. there is no recognized “official” Indian-version of the instrument? Is that correct?
Reply: We will include studies that were validated to be used in India among type-2 diabetes population. We agree, that the statement was confusing hence it has been rephrased. Please refer lines 157-60.
13. (line 177, Comment JC13) This does not appear to follow the search strategy outlined in the COSMIN manual. You do not refer to the PROMs filter developed by Terwee et al, 2009.
Reply: We agree with the comment. At the time of submission of the protocol, the search strategy did not include search filters suggested by Terwee et al., 2009. We are currently working with the information scientist to finalize the search using filters suggested by Terwee et a., 2009. Final search strategy will be shared in the completed review. However, in the current protocol, we have reported the search concepts and filters suggested by Terwee et al., 2009. Please refer to lines 166-76 and

table 1.
14. (line 217, Comment JC14) Is this correct? The COSMIN manual has
Content validity – development study 35 Qs, other content
validity studies 31 Qs
Structural validity – 4 Qs
Internal consistency – 4 Qs
Cross-cultural validity/measurement invariance – 3 Qs
Reliability – 8 Qs
Measurement error – 5 Qs
Criterion validity – 3 Qs
Construct validity – 6 Qs
Responsiveness – 12 Qs
Perhaps the manuals should be included as part of the references to demonstrate how you will assess the psychometric properties?
The heading "Quality appraisal of included studies" is also confusing to the reader, particularly if they are unfamiliar with the COSMIN approach. One of the benefits of COSMIN is that it incorporates assessment of the study quality as part of the rating/scoring of the PROMs that are being assessed. The modified GRADE approach is applied (see pages 32-36 of the COSMIN manual).
Reply: Thank you for making this clear. Yes we agree with the comment. After a thorough discussion, team members have decided to restrict the assessment of risk of bias for content validity and internal structure. We have reported the reasons in main manuscript file. Considering the comment, we now have added few more references and manual, which will help in understanding the methodology we are intending to follow. Please refer to section 'Risk of bias assessment of included studies' starting from line no. 203.
Also, we agree with the comment regarding heading and therefore have edited it to "Risk of bias assessment of included studies", line no. 203.
15. (line 237, Comment JC15) I think you are potentially underplaying this. At the end of the review you will be able to provide evidence of the psychometric properties for each of the PRO instruments included in the study. That is a rating for each property. COSMIN methods also dictate that you should be able to make a recommendation based upon the results of the review (see COSMIN checklist for COSMIN review methods).
Reply: Thank you for the suggestion. We now have addressed these points in the manuscript file. Please refer lines 232-37.

VERSION 3 – REVIEW

REVIEWER	Carlton, Jill University of Sheffield, School of Health and Related Research
REVIEW RETURNED	15-Mar-2021

GENERAL COMMENTS	The authors have done an excellent job in revising the manuscript. It is now much easier to understand, and does reflect the processes involved for a systematic review of PROs using the COSMIN methodology. The results of the review will be of interest to the field of diabetes research.